VerifierShift Restricted Evidence for Changed Agent Workflows

SAURAV K. ARYAL, Howard University, USA

Downstream agent verifiers can retain complete tool responses or deliberately restrict themselves to typed predicate facts. The latter reduces raw-field disclosure but can require another observation when a policy changes. We study this tradeoff through a frozen replay of public tau2 retail tool methods. Of 40 provider-test tasks, 31 contain an executable first order-write trace; nine controlled edits yield 279 repeated task/edit pairs and 1,395 method decisions. Always-fresh, dependency-aware, sufficient-statistic, semantic-entailment and compiled-capsule methods all resolve 217 supported cases correctly and review 62 unsupported cases. Dependency-aware reuse, Z3 and compiled capsules each require 93 verification reads; retaining sufficient statistics requires 62, versus 217 for always-fresh verification. Mean retention is 505 bytes for capsules, 414 for projection and 1,394 for raw responses, including common provenance metadata. Projection therefore wins reads and bytes. Capsules retain four predicate bits without raw source scalars, but still reveal eligibility information. These results establish a restricted interface and explicit disclosure-versus-refresh tradeoff, not a superior caching algorithm, formal privacy, full workflow correctness or model-agent performance. Future work will determine whether realistic downstream disclosure constraints justify the extra refresh cost.

Additional Key Words and Phrases: agent verification, typed evidence, evidence minimization, policy drift, selective refresh

1 Introduction and Background

A retail tool can return an order, shipping address, item identifiers and payment information when a downstream verifier needs only whether an action is currently permitted. Retaining the full response makes later questions easier to answer, while increasing the source information shared with the verifier. Retaining a small set of predicate outcomes changes that tradeoff: a later requirement may need a source fact that the initial capsule discarded. We ask when this restriction preserves useful decisions, and what additional observations it requires.

Evidence reuse itself is established. Safe and Sufficient Caching validates access and freshness before reusing retrieval evidence [6]. Recent assurance-evidence research characterizes dependency validity and the risk of undeclared dependencies [3]. ActiveEvidence studies decision-focused evidence acquisition with model-relative certificates [2]. Recent VeriCUA work treats verification as active exploration or learned checking [5, 7]. These precedents rule out novelty claims based only on hashes, freshness gates, selective reads or Boolean compilation. Our contribution is a small reproducible interface study that makes strong baseline parity and restricted-disclosure costs observable.

The experiment replays genuine public tool methods and data from a pinned tau2-bench revision [4]. It is a deliberately narrow verification fragment, not an actor benchmark. The repository's earlier authored-schema/model-judge study remains a separate historical component; its 168 model calls are not counted here. Code, frozen results, audit records, and the manuscript package are available at https://github.com/Saurav-K-Aryal/verifiershift-pilot.

2 Methods

2.1 Public workflow replay and supported obligations

We pin Sierra Research's MIT-licensed tau2-bench repository to commit `da0835f6` (full revision in the artifact). Provider training tasks support adapter development. Confirmation uses all 40 provider-test task identifiers after code, protocol and audit hashes are frozen. For each task, the expected action trace exercises the unmodified public tools until the first order-writing action. The original native write and a restricted status postcondition are checked separately from subsequent evidence-reuse decisions. Tasks with no order write or native errors remain in the inclusion ledger.

Author's Contact Information: Saurav K. Aryal, Howard University, Washington, DC, USA.

The supported fragment checks four facts: a trusted caller context is present, the order belongs to that caller, its status is appropriate for the action, and the request has the required item/reason/address scope. Every task uses the same declared post-authenticated-session assumption: the caller principal is fixed to the original fixture owner. Expected traces contain an authentication lookup in only 16 of the 31 eligible tasks. We do not infer observed actor authentication from this assumption. Dialogue confirmation, payment settlement, unrestricted free text and complete natural-language policy compliance are outside the fragment.

Each eligible task receives nine authored interventions: reorder the same conjunction; remove the request obligation; negate the required status; introduce an amount cap of 500; change the actual source status; change the actual owner; change an irrelevant shipping ZIP field; introduce unsupported external risk; and introduce unknown tool semantics. Field-change notices identify changed paths without giving their new values. These interventions are controlled wrapper edits, not a native tau2 drift benchmark or an estimate of natural change frequencies.

2.2 Evidence representation and fair controls

A capsule contains four predicate outcomes, declared dependencies, record version and digests binding the source record, policy and adapter. An irrelevant source edit can preserve evidence, while a relevant edit invalidates the affected facts. A new amount-cap predicate is not in the original capsule and requires a read. Unsupported requirements or tool semantics yield review. Hashes bind trusted local records; they are not remote authentication or distributed freshness guarantees.

Five methods share the same notifications, source access and unit read price. Always-fresh verification reads the order for every supported case. Ordinary dependency/validity reuse uses short-circuit Boolean reasoning over surviving facts. A sufficient-statistic projection additionally retains policy-relevant raw fields, including the amount needed by the new cap. Semantic entailment uses Z3 [1] over the same surviving capsule facts. Compiled transport enumerates the same five-atom finite domain into a lookup representation. It is expected to match semantic reasoning, rather than outperform it through access to extra information. The projection intentionally retains more raw facts and is an explicit disclosure control.

Retention measures include identical common provenance metadata. We report serialized bytes, raw source scalar counts and predicate bits separately. Zero raw fields does not mean zero information leakage: predicate outcomes can disclose eligibility and support inference. No formal privacy definition or privacy guarantee is tested.

2.3 Freezing and verification

Before confirmation, development exposed an inadequate short-circuit baseline, a projection amount-overwrite error and incomplete authentication witnesses in expected traces. The first attempt is preserved; the corrected protocol and uniformly declared caller-context assumption were frozen before test execution. A separately implemented arithmetic/Boolean oracle recomputes the controlled truth directly from the public raw fixtures. A second automated replay compares all scientific decision/read fields. Experiment implementation and analysis used Codex assistance under the author's research direction.

The units are the original tasks and source records. The 31 eligible tasks contain 28 orders and 25 callers; 15 confirmation tasks share 13 orders with development. This is task-held-out interface and arithmetic feasibility, not source-disjoint generalization. Nine repeated edits on a task are not nine independent observations. We report exact finite counts without a population-level reliability interval. Runtime is a local microbenchmark and does not establish service latency or deployment throughput.

Table 1. Frozen confirmation across 31 task groups and 279 repeated task/edit pairs per method. Review covers unsupported cases.

Method	Correct supported	Review	Native reads
Always fresh	217/217	62/62	217
Dependency and validity	217/217	62/62	93
Sufficient-statistic projection	217/217	62/62	62
Semantic Z3	217/217	62/62	93
Compiled capsule	217/217	62/62	93

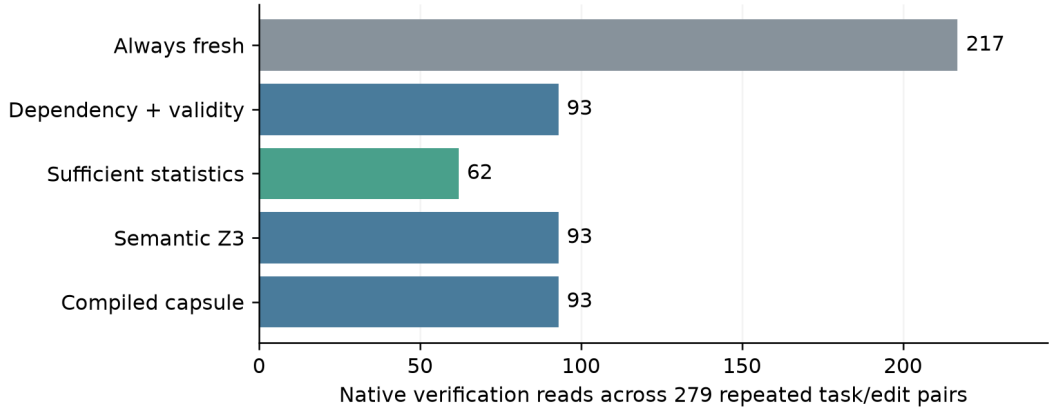


Fig. 1. Native read counts on the same 279 repeated task/edit pairs. Unsupported cases return review without a read. Strong dependency, solver and compiled controls match; sufficient statistics require fewer reads.

3 Results

3.1 Coverage and native replay

Thirty-one of the 40 test tasks have an executable first order write. Four have no order write, four fail a preceding tool call, and one fails the native write. All 31 eligible native writes and restricted status postconditions pass. The eligible actions cover cancellation (3), exchange (8), address modification (6), item modification (8), payment modification (1), and return (5). These are gold-trace tool replays, not successful agent episodes.

The nine edits produce 279 task/edit pairs. Each method correctly resolves all 217 supported cases and returns review for all 62 cases with unsupported external risk or unknown tool semantics. There are no incorrect accepts within this controlled fragment. Identical outcomes across the five controls show consistency on the authored cases, rather than establishing a general safety rate.

3.2 Read and retention tradeoffs

Dependency-aware reuse, semantic entailment and compiled capsules have exactly the same decisions and reads for all 279 pairs. Each makes 93 verification reads, compared with 217 for always-fresh verification. The retained projection makes 62 reads because it can answer the new amount-cap requirement without another observation. The three methods that retain only the initial four facts must fetch that missing fact. Figure 1 makes this parity explicit.

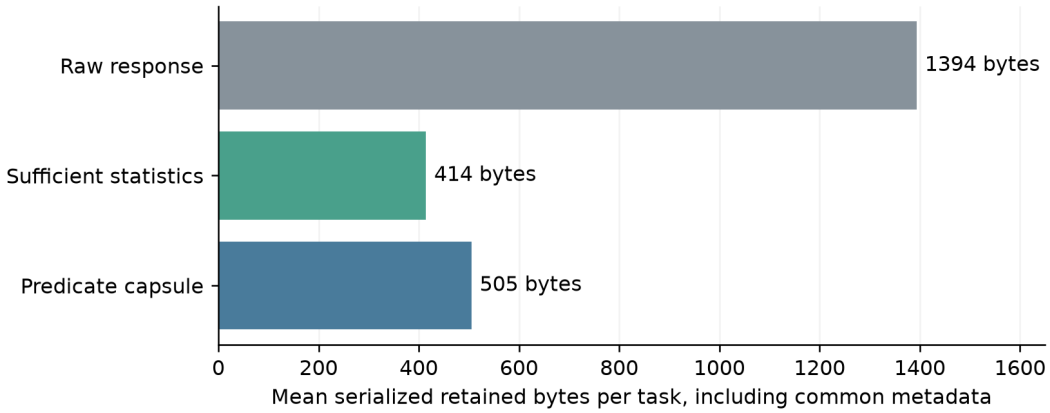


Fig. 2. Mean serialized retention across the 31 eligible tasks. Capsule facts narrow raw-field disclosure, but the sufficient-statistic projection is smaller. The measures are not estimates of information-theoretic privacy.

Mean serialized retention is 1,393.6 bytes for the raw response, 414.1 for the projection and 505.0 for the capsule, including common metadata (Figure 2). The projection retains approximately 6.3 raw source scalars, compared with four predicate bits and no raw source scalars in the capsule. The raw response contains approximately 44.2 source scalars. Predicate dependency metadata makes the capsule larger than the compact projection in this implementation. It cannot be recommended on byte minimization or refresh cost alone.

The separate raw-fixture audit reports zero discrepancies across all 1,395 method decisions. Full automated replay reproduces every decision, source notice, known fact and read count. These checks improve reproducibility of the finite construction. They do not independently validate the completeness of the four-predicate contract for real retail workflows, dishonest change notices or concurrent distributed sources.

4 Future Work

The next study should determine whether an actual downstream disclosure restriction justifies predicate capsules despite the projection's measured advantage. A larger public corpus can compare decision coverage, retained source facts and refresh cost across explicit disclosure budgets. Shared orders, callers and policy families should remain in one partition; naturally occurring changes and authored interventions should be reported separately. A second public workflow adapter and independently audited typed specifications can test the limits of the supported fragment without treating missing dialogue or inaccessible state as known truth.

Strong controls remain essential: ordinary dependency/validity reasoning, sufficient-statistic projection, always-fresh reads and semantic entailment should share notifications and costs. If projection dominates for a use case, the reference toolkit should recommend it. Unsupported constructs and untrusted notifications need separate failure tests. Bounded model-agent experiments can follow adapter validation, but native task evaluators must score actor success separately from evidence-obligation coverage. The intended output is a measured disclosure/coverage/read-cost frontier and reusable source/verifier interface, not a universal verification or privacy guarantee.

5 Conclusion

The controlled public-tool replay demonstrates a reproducible typed-evidence interface and a concrete limitation of restricted disclosure. Capsules match strong dependency and semantic controls, while retained sufficient statistics use fewer reads and bytes. The useful research question is when deliberate limits on downstream source information justify that cost. The current evidence supports studying that boundary and openly preserving the cases where the simpler projection is preferable.

References

[1] Leonardo de Moura and Nikolaj Bjørner. 2008. Z3: An Efficient SMT Solver. In *Tools and Algorithms for the Construction and Analysis of Systems*. Springer, 337–340. doi:10.1007/978-3-540-78800-3_24

[2] Miaobo Hu, Shuhao Hu, Xiaobo Guo, Xin Wang, Bokun Wang, Peng Zhang, and Daren Zha. 2026. ActiveEvidence: Decision-Focused Evidence Graphs with Verifiable Acquisition Gaps. https://arxiv.org/abs/2610.20532v2 arXiv:2610.20532v2.

[3] Atanu Mukhopadhyay and Shravya Reddy Nalla. 2026. Reusing Assurance Evidence in AI Pipelines: Validity Conditions, a Fail-Closed Release Gate, and a Replay of 714 Public Data Version Control Pipelines. https://arxiv.org/abs/2610.27672v1 arXiv:2610.27672v1.

[4] Sierra Research. 2026. tau2-bench public retail tools, data models and fixtures. https://github.com/sierra-research/tau2-bench/tree/da0835f6c67baef58bdda18f2a3b037dfb3a1754 MIT license; commit da0835f6c67baef58bdda18f2a3b037dfb3a1754; accessed November 1, 2026.

[5] Qiushi Sun, Da Yu, Alexander Knop, Amer Sinha, Lingpeng Kong, Ben Kao, Ravi Kumar, and Pritish Kamath. 2026. VeriCUA: Verification as Agentic Exploration for Computer-Using Agents. https://arxiv.org/abs/2610.40578v1 Work in progress, arXiv:2610.40578v1.

[6] Akshay Verma, Siddharth Pillai, Sambit Das, and Deepak Gupta. 2026. Safe and Sufficient Caching: Validated Evidence Reuse for Production Retrieval-Augmented Generation. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, 1272–1279. https://aclanthology.org/2026.emnlp-industry.75/

[7] Zhizhou Wang, Yaxi Lu, Zhong Zhang, Zikun Hu, Xin Cong, Yankai Lin, and Fangming Liu. 2026. VeriCUA: Learning Proactive Verification for Reliable GUI Agent Rewards. https://arxiv.org/abs/2610.26469v1 arXiv:2610.26469v1.